

Experiment Report

Anomaly Detection with MLflow Experiment Tracking

Objective

The objective of this project was to build an anomaly detection system capable of identifying fraudulent transactions from a highly imbalanced dataset while maintaining a reproducible machine learning experimentation workflow using MLflow.

Dataset Description

Dataset Used:
Fraud Detection Dataset

Source:
<https://www.kaggle.com/datasets/kartik2112/fraud-detection>

The dataset contains transaction records labeled as either:

- Class 0: Legitimate Transaction
- Class 1: Fraudulent Transaction

The dataset exhibits severe class imbalance, making anomaly detection a challenging machine learning problem.

Data Preprocessing

The following preprocessing steps were performed:

1. Dataset loaded using Pandas.
2. Class distribution inspected.
3. Sampled 200,000 records for efficient experimentation.
4. Removed irrelevant columns.
5. Encoded categorical features using LabelEncoder.
6. Performed a stratified 70:30 train-test split.

The test set remained untouched until final evaluation.

Class Imbalance Handling

SMOTE (Synthetic Minority Oversampling Technique) was applied only to the training data.

Purpose:

- Increase minority class representation.
- Improve fraud detection capability.
- Reduce model bias toward the majority class.

Class distribution before and after SMOTE was recorded and verified.

Experiments Conducted

Experiment 1: Logistic Regression

Parameters:

- $C = 1$
- solver = liblinear

Result:

The model failed to identify fraudulent transactions and achieved zero recall for class 1.

Experiment 2: Random Forest

Parameters:

- n_estimators = 30
- max_depth = 3

Result:

The model exhibited behavior similar to Logistic Regression and failed to detect minority class samples effectively.

Experiment 3: XGBoost

Parameters:

- `eval_metric = logloss`

Result:

XGBoost significantly improved fraud detection performance and achieved a class 1 recall of approximately 59%.

Experiment 4: XGBoost With SMOTE

Parameters:

- `eval_metric = logloss`
- SMOTE applied on training data

Result:

This configuration achieved the highest fraud detection capability and significantly improved minority class recall.

Experiment Results

Model	Accuracy	Recall Class 1	F1 Macro
Logistic Regression	0.99405	0.00000	0.49851
Random Forest	0.99405	0.00000	0.49851
XGBoost	0.99630	0.59384	0.82724
XGBoost With SMOTE	0.99110	0.75070	0.74822

MLflow Experiment Tracking

All experiments were tracked within a single MLflow experiment.

For each run, the following were logged:

Parameters

- Model hyperparameters
- Sampling strategy

Metrics

- accuracy
- recall_class_0
- recall_class_1
- f1_score_macro

Artifacts

- Trained model files

Model Registry Workflow

Best Model: XGBoost With SMOTE

Registered Model Name: anomaly-detector-xgb-smote

Alias: challenger

Production Model: anomaly-detection-prod

Alias: champion

The production model was successfully loaded from the MLflow registry and used for inference on test samples.

Conclusion

The experiments demonstrated the importance of handling class imbalance in anomaly detection problems.

While XGBoost achieved the highest overall accuracy, XGBoost with SMOTE significantly improved fraud detection performance by increasing recall for the minority class.

Therefore, XGBoost with SMOTE was selected as the production model and deployed through the MLflow Model Registry.

The project successfully implemented a complete machine learning lifecycle including experimentation, tracking, model registration, alias management, and production inference.